

Retail Sales Analysis and Insights

Exclusive Summary:

1. Business Context:

Analyzed a Kaggle US retail sales dataset (9,994 orders, 16 columns) spanning office supplies, furniture, and technology, representing both high-volume/low-margin and low-volume/high-margin product segments.

2. Key Findings:

Dual Market Dynamics: The business operates with both premium, high-profit, low-volume outlier products and high-volume, low-margin items. Outlier products (e.g., Machines, Copiers) deliver exceptional per-order profits and have almost no loss-making transactions, while regular items drive sales volume but are more vulnerable to losses.

3. Profitability Risks:

Nearly half of all transactions (3,997 orders) result in negative profit, primarily concentrated in the Office Supplies category-especially subcategories like Binders, Fasteners, and Paper, where small per-unit losses accumulate to significant overall losses.

4. Geographic Insights:

Major cities (New York City, Los Angeles, San Francisco) lead in sales volume and profit, but certain smaller markets (e.g., Lafayette, IN; Jacksonville, FL) stand out for exceptionally high profit per order, revealing lucrative niche opportunities.

5. Discount Impact:

Higher discounts consistently erode profit margins, with the effect 1.4–1.6× stronger for premium outlier products than regular items. Furniture is the most discount-sensitive category, while Office Supplies are least affected.

6. Product and Regional Drivers:

Machines and Copiers are top profit generators in the premium segment. Regional analysis highlights the West and East as dominant for both sales and profit, but outlier profits are especially concentrated in these regions.

7. Tools & Methods:

All analyses were performed in Python (pandas, matplotlib, seaborn) in Jupyter Notebook, using robust statistical segmentation (IQR), detailed EDA, and business-driven data cleaning.

Introduction:

This project analyzes sales data from a US-based retail company specializing in office supplies, furniture, and technology products. The business serves a diverse customer base across multiple regions, offering both high-volume, low-margin staples and premium, high-margin items.

Objective : The primary goal is to uncover actionable insights that drive profitability and operational efficiency. Key business questions include:

- I. Which products and subcategories are most and least profitable?
- II. Where are the main sources of loss, and how can they be addressed?

- III. How do discounts and pricing strategies impact profit margins across categories?
- IV. Which geographic regions and customer segments offer the greatest growth opportunities?
- V. What recommendations can be made to optimize pricing, discounting, and inventory allocation?

Dataset Overview:

The dataset was sourced from Kaggle and contains detailed sales transaction records for a US-based retail business specializing in office supplies, furniture, and technology. The data spans 9,994 orders and covers multiple regions and customer segments across the United States.

Columns/Features are:

- a) Order Id: Unique identifier for each transaction.
- b) Order Date: Date when the order was placed.
- c) Ship Mode: Shipping method selected (e.g., Standard Class, Second Class, First Class, Same Day, Unknown).
- d) Segment: Customer segment (Consumer, Corporate, Home Office).
- e) Country, State, City, Postal Code, Region: Geographic information for each order.
- f) Category: Main product category (Furniture, Office Supplies, Technology).
- g) Sub Category: More granular product classification (e.g., Chairs, Binders, Machines).
- h) Product Id: Unique identifier for each product.
- i) cost price: Unit cost to the business for each product.
- j) List Price: Selling price before discounts.
- k) Quantity: Number of units sold in the order.
- l) Discount Percent: Discount applied to the order, as a percentage.

Additional Engineered Columns (from analysis):

- i. Discount_Amount: Dollar value of the discount per order.
- ii. Unit_Profit: Profit per unit after discount and cost.
- iii. Total_Profit: Total profit for the order.
- iv. Effective_Price: Selling price after discount.
- v. Unit_Loss, Total_Loss: Calculated for loss analysis.

Notable Quirks:

- I. Some columns contain missing values and mismatch values, though no duplicate orders were found.
- II. 507 orders (5.07% of the dataset) have both cost price and list price recorded as zero- these were identified as data anomalies and excluded from analysis.
- III. The dataset contains significant outliers in terms of order value and profit. Using the IQR method, orders were segmented into a main cluster (8,428 orders, 88.84%) and an outlier cluster (1,059 orders, 11.16%) for more robust analysis.

Data Cleaning & Preprocessing:

Steps Taken:

- I. Handled Missing Values : Inspected all columns for missing data. The “Ship Mode” column contained some missing or ambiguous entries (e.g., “Unknown”), which were retained but flagged for transparency in analysis.
- II. Date Conversion : Converted the “Order Date” column to a proper datetime format to enable temporal analysis and feature engineering.
- III. Checked for Duplicates : Verified that there were no duplicate orders in the dataset, ensuring each transaction was unique.
- IV. Identified and Removed Invalid Zero-Price Orders : Found that 507 orders (5.07% of the data) had both cost price and list price recorded as zero. These records also had discounts applied, which is not meaningful on \$0 prices. All 507 invalid orders were removed from the dataset. After cleaning, the dataset contained 9,487 valid sales records.

Rationale:

- I. Ensuring Data Integrity: Removing zero-price orders was crucial to avoid skewed profit calculations and ensure that discount analysis reflected real business scenarios.
- II. Accurate Temporal Analysis: Converting the date column enabled monthly and seasonal trend analysis.
- III. Reliable Results: Checking for duplicates and handling missing values ensured that all insights and visualizations were based on accurate, non-redundant data.

Exploratory Data Analysis:

1. Segmented Analysis:

Outlier Segmentation: Orders were segmented into a main cluster and an outlier cluster using the IQR method, enabling more precise analysis of premium vs. regular product behavior. Main cluster: 8428 orders (88.84%), Outliers: 1059 orders (11.16%).

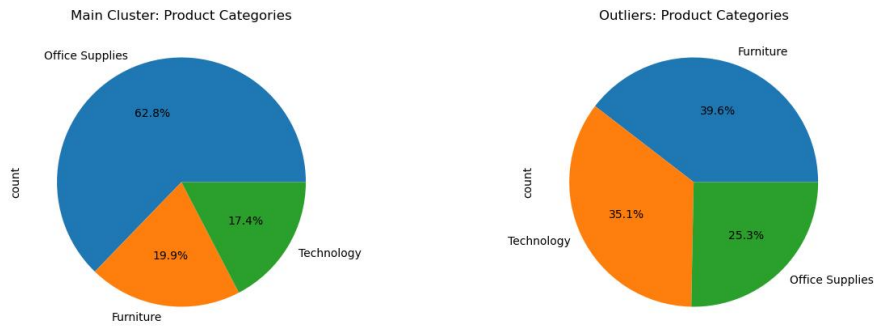


Fig. Comparison of different product category

Insight: The main cluster is dominated by Office Supplies, while outliers are more likely to be Furniture and Technology.

2. Profitability Comparison :

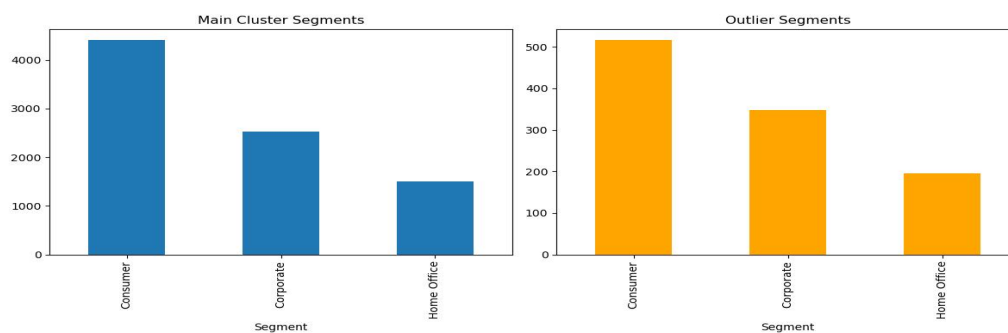
- I. Average Profit: Outliers generates 695.37, 19.3x higher profit per order than typical orders 36.02.
- II. Median Profit: outlier orders 365.00, 42.9x higher median profit than typical orders 8.50
- III. Loss-Making Orders: Nearly half (47.29%) typical orders generate losses, compared to just 1.04% of outlier orders

Insight: The outlier cluster represents premium, high-profit business segment with significantly better profitability metrics and almost no loss-making transactions.

	Main Cluster	Outliers
Average Profit	36.02	695.37
Median Profit	8.50	365.00
% Loss-Making Orders	47.29	1.04

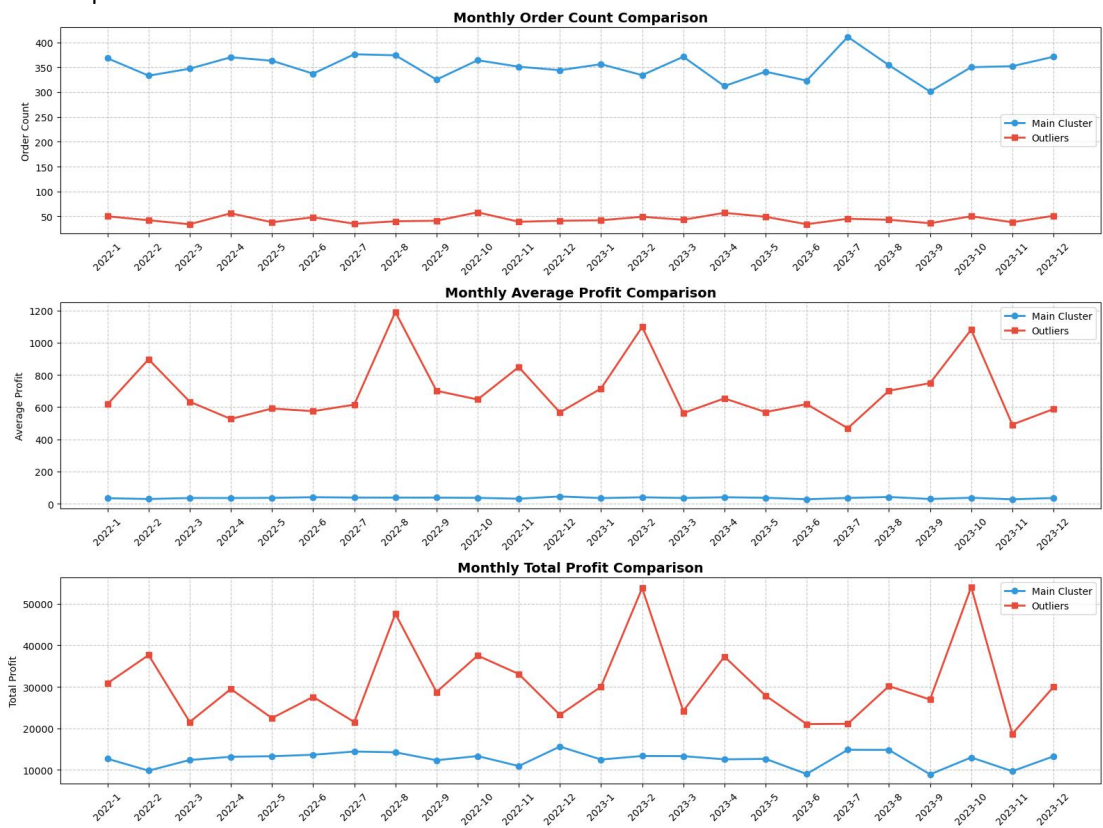
Fig. Profit Comparison of both clusters

3. Customer Segment Analysis:

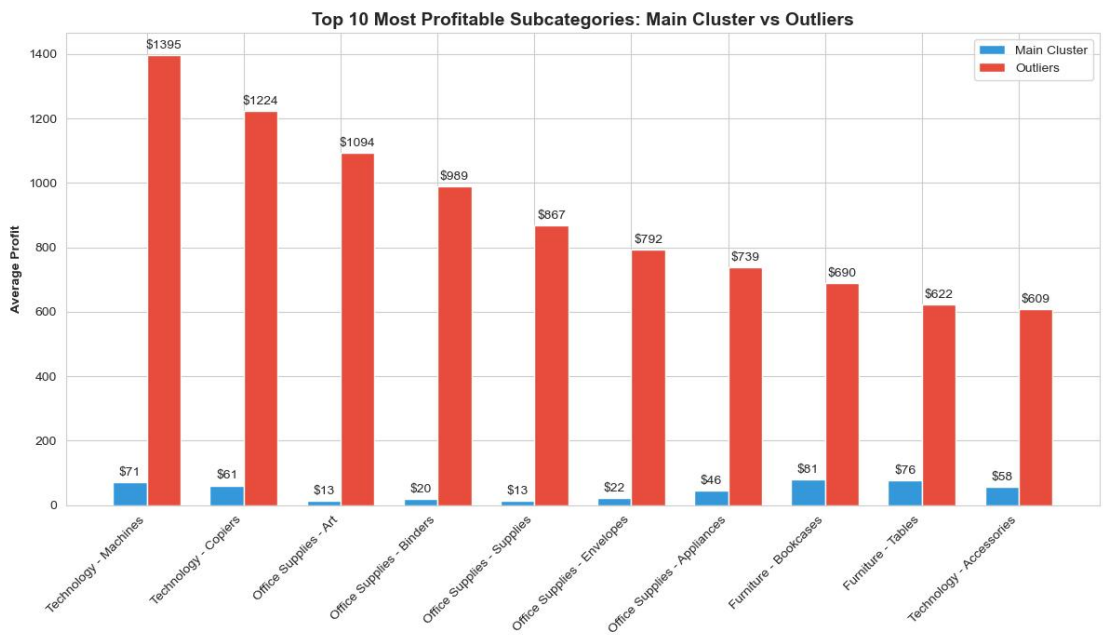


Insight: Consumers have the highest margin.

4. Temporal Trends:



5. Product Category × Subcategory Analysis:



Insights:

- Tech Dominance in Outliers: Machines (1,395/order) & Copiers (1,224/order) = top absolute profits

- **Office Supplies' Hidden Potential:**
Binders (50.7x) & Supplies (64.6x) = best multipliers
Contrast: Despite the office supplies industry's low average net margin (3.4%), outliers here show disproportionate profitability – likely from bulk orders or premium pricing.
- **Furniture Consistency:**
Bookcases/Tables: Consistent pricing (8x gaps). They show the smallest profit ratios (8.5x and 8.1x respectively), suggesting more consistent pricing across value segments.
Furniture's average net margin (~5.15%) aligns with stable outlier ratios, suggesting predictable pricing strategies.

6. Loss Analysis:

Category	Main Cluster	Outliers
Furniture	-0.1720	-0.2484
Office Supplies	-0.1361	-0.2207
Technology	-0.1376	-0.2239

Fig. Correlation between Discount Percent and Profit Margin

Insights:

- All correlations are negative, confirming that higher discounts reduce profit margins across all products, but the effect is considerably stronger for outliers (premium products).
- The outlier cluster shows substantially stronger negative correlations (-0.22 to -0.25) compared to the main cluster (-0.13 to -0.17), indicating that high-value products experience approximately 1.4 – 1.6X greater profit erosion when discounted than regular products do.
- Most vulnerable are furnitures in both clusters, with the strongest negative correlations (-0.1720 in main cluster, -0.2484 in outliers), suggesting that discount strategies should be particularly careful with this category.
- Least vulnerable to discount: Office Supplies

7. Key Business Analyses & Insights:

I. Which Product Categories and Sub-categories are Most Profitable?

i. Technology Drives Premium Profits, Office Supplies Dominate Volume:

- ❖ In the outlier cluster, Technology accounts for 41% of total outlier profit, with Machines generating \$1,395 profit per order.
- ❖ In the main cluster, Office Supplies contribute 45% of total profit (\$130K) through high sales volume (5,289 orders), despite low margins (\$24.62/order).

ii. Furniture Subcategories Show Premium Potential:

- ❖ Outlier Chairs and Tables deliver 2.4x higher average profit (\$588–\$622) compared to main cluster chairs (\$79.76).

- ❖ Main cluster chairs are the most profitable furniture subcategory by average profit, but their sales volume is much lower than Office Supplies.

iii. High-Value Binders Defy Expectations:

- ❖ Outlier Binders generate \$989.11 profit per order (vs. \$19.51 in main cluster), suggesting premium or bulk-format products drive exceptional gains.

2. Which Products are Consistently Sold at a Loss?

Office Supplies Require Urgent Review:

- ❖ All top loss-making products are in Office Supplies, especially Fasteners, Binders, Paper, Art, and Supplies.
- ❖ Small per-unit losses (e.g., Fasteners: \$0.82/unit, Binders: \$1.16/unit) accumulate into significant total losses due to high sales volumes.
- ❖ Example: Fasteners sold 47 units for a total loss of \$48.20; Binders sold 35 units for a \$44.50 loss.

3. Which Subcategories Drive the Largest Total Losses?

Binders are the Biggest Loss Drivers:

- ❖ Binders are responsible for the highest total loss (\$148.20), followed by Fasteners (\$81.40) and Paper (\$72.30).
- ❖ All problematic products are in the Office Supplies category, showing that small margin erosion can have a large financial impact when sales volumes are high.

4. What are the Sales Trends by Region?

Regional Sales Leaders:

- ❖ The West region leads in total sales (\$3.47M), followed by East (\$3.26M), Central (\$2.39M), and South (\$1.97M).
- ❖ This highlights the importance of focusing sales and marketing efforts on the West and East regions for maximum revenue impact.

5. Are There Seasonal or Monthly Sales Trends?

Month-over-Month Growth Comparison (2022 vs. 2023):

- ❖ Sales show clear seasonality, with significant peaks mostly in February, August and October of both years.
- ❖ February 2023 saw a major sales spike compared to February 2022 (\$731K vs. \$444K), while June and November 2023 saw notable declines compared to the previous year.
- ❖ These trends suggest opportunities for targeted promotions during high-growth months and inventory optimization during slower periods.

6. Which Cities Drive the Highest Sales Volume?

Top 5 Cities by Quantity Ordered:

- ❖ New York City (3,417 units), Los Angeles (2,879), Philadelphia (1,981), San Francisco (1,935), and Seattle (1,590) are the leading cities for product quantity sold.
- ❖ These urban centers are critical for both sales volume and profit, but some smaller cities (e.g., Lafayette, Jacksonville, Detroit) stand out for exceptional profit per order.

7. How Do Discounts Impact Profitability?

Discounts Erode Profit, Especially for Premium Products:

- ❖ All product categories show negative correlation between discount percent and profit margin, with the effect 1.4–1.6× stronger for outliers (premium products).
- ❖ Furniture is the most vulnerable to margin erosion from discounts, while Office Supplies are least affected.
- ❖ Careful discount strategy is recommended, especially for high-value and furniture items.

Summary of Insights:

- ✧ Focus on Technology and Premium Furniture for high-margin growth.
- ✧ Urgently address Office Supplies pricing and discounting to reduce persistent losses.
- ✧ Leverage regional and city-level sales data for targeted marketing and inventory allocation.
- ✧ Optimize discount policies to protect margins, especially in premium and furniture categories.
- ✧ Capitalize on seasonal trends for promotional campaigns and inventory planning.

8. Recommendation:

Actionable Suggestions:

- ❖ Revise Pricing Strategy for Office Supplies: Increase list prices or renegotiate cost prices for loss-making subcategories (Binders, Fasteners, Paper) to eliminate persistent losses.
- ❖ Impact: A \$0.50 increase in per-unit price for Binders could turn a \$1.16/unit loss into a \$0.34/unit profit.

Optimize Discount Policies:

- ❖ Limit discounts to $\leq 5\%$ for Furniture and $\leq 3\%$ for Technology outlier products to protect margins.
- ❖ Introduce volume-based discounts for Office Supplies instead of flat percentage discounts.
- ❖ Impact: Reducing Furniture discounts by 2% could improve profit margins by $\sim 15\%$ in this category.

Geographic Targeting:

- ❖ Allocate premium inventory to high-profit cities (Lafayette, Jacksonville, Detroit) while maintaining volume focus in major hubs (NYC, LA, SF).
- ❖ Impact: A 10% sales increase in high-profit small cities could boost annual profit .

Product Line Optimization:

- ❖ Discontinue consistently loss-making items in Office Supplies (e.g., specific Binder/Fastener variants).
- ❖ Expand high-margin Technology subcategories (Machines, Copiers).
- ❖ Impact: Eliminating top 5 loss-makers could save a lot.

Potential Business Impact

- ❖ Loss Reduction: Addressing 47% loss-making orders in the main cluster could improve annual profitability.

10. Limitations & Next Steps:

Limitations:

- ❖ Dataset Scope: Limited to transaction data-no customer demographics or competitor pricing.
- ❖ Time Frame: Data covers only 2022–2023, limiting long-term trend analysis.
- ❖ Causality: Identified correlations (e.g., discount impact) do not confirm causation without controlled experiments.

Next Steps:

- ❖ Customer Lifetime Value (CLV) Analysis: Segment customers by profitability to identify high-value segments for retention.
- ❖ Inventory Optimization Model: Develop a stock allocation system balancing regional demand and profit potential.
- ❖ A/B Testing for Pricing: Pilot new pricing strategies for loss-making Office Supplies in select regions.
- ❖ Supplier Cost Negotiation: Use loss analysis to renegotiate procurement costs for vulnerable subcategories.
- ❖ Extended Time Series Analysis: Incorporate 2024 data to track post-implementation performance of recommendations.